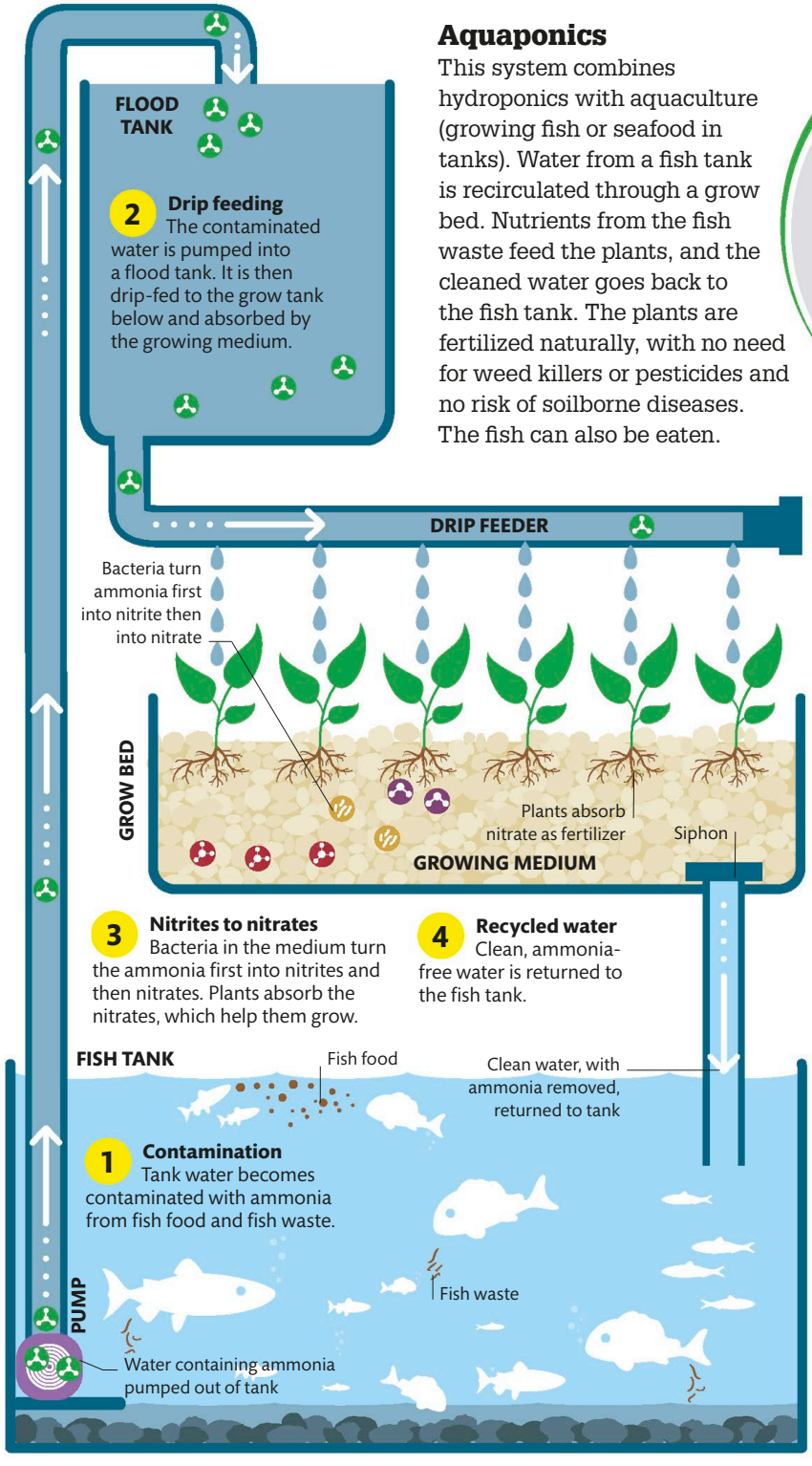
Aquaponics

This system combines hydroponics with aquaculture (growing fish or seafood in tanks). Water from a fish tank is recirculated through a grow bed. Nutrients from the fish waste feed the plants, and the cleaned water goes back to the fish tank. The plants are fertilized naturally, with no need for weed killers or pesticides and no risk of soilborne diseases. The fish can also be eaten.

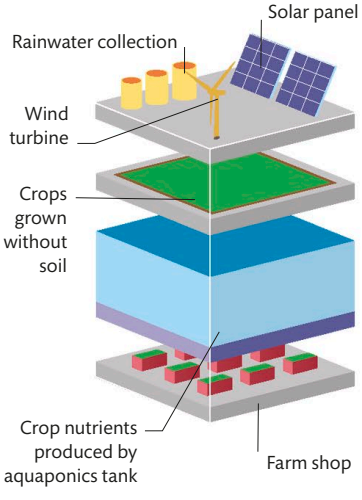
HOW MUCH SPACE DOES HYDROPONIC FARMING SAVE?

Farmers can plant between 4 and 10 times the number of plants hydroponically in the same space as a traditional farm.



VERTICAL FARMING

Urban farms may one day feature soilless systems in skyscrapers. Crops could be grown in vertical shelf systems or on lightweight decks. Robots would tend and harvest the crops, while sensors would monitor crop growth.



KEY

- Ammonia
- Bacteria
- Nitrite
- Nitrate